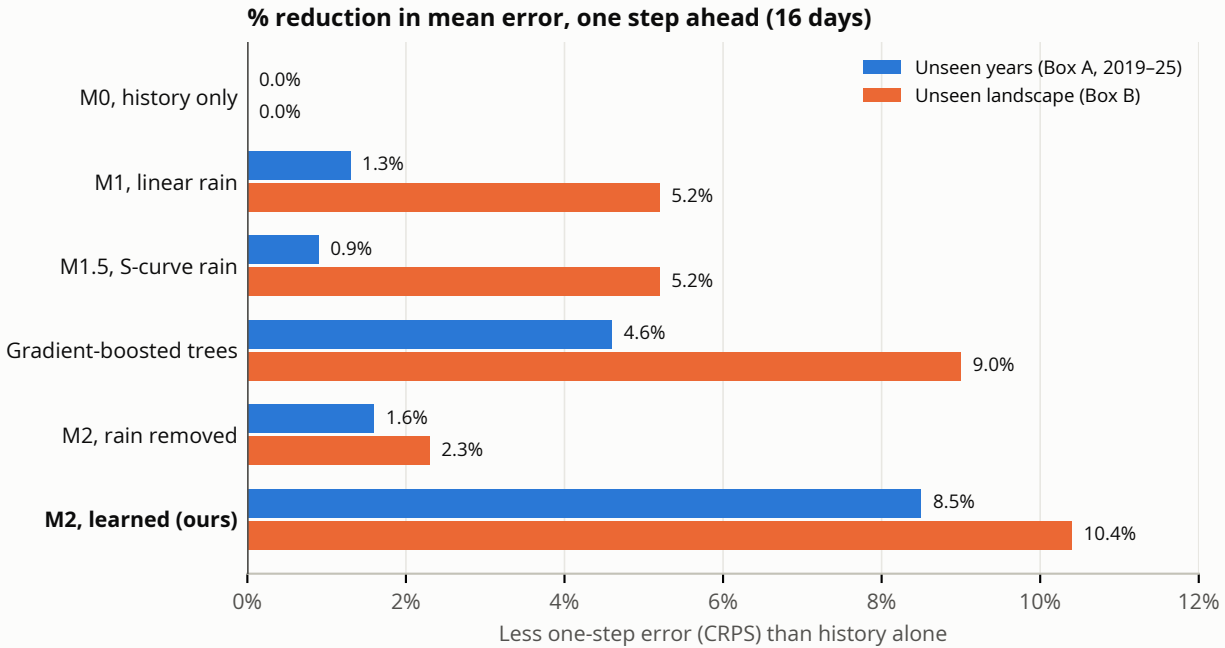
% reduction in mean error, one step ahead (16 days)
Unseen years (Box A, 2019–25)
Unseen landscape (Box B)
M0, history only
0.0%
0.0%
M1, linear rain
1.3%
5.2%
M1.5, S-curve rain
0.9%
5.2%
Gradient-boosted trees
4.6%
9.0%
M2, rain removed
1.6%
2.3%
M2, learned (ours)
8.5%
10.4%
0%
2%
4%
6%
8%
10%
12%
Less one-step error (CRPS) than history alone